

YIFAN MEGAN ZHAO

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EDUCATION

University of Pennsylvania
Ph.D. in Physics

Philadelphia, PA, USA
Aug 2024 - May 2029 (expected)

University of Chicago
B.S. in Astrophysics with honors; minor in Theater & Performance Studies
Honors thesis: "Convolutional Neural Network for Type Ia Supernova Distance Estimation"

Chicago, IL, USA
Sep 2020 - June 2024

PUBLICATIONS

- **M. Zhao**, P. Swierc, A. Ćiprijanović, B. Nord, "Domain Adaptation for Measurements of Strong Gravitational Lenses", accepted by NeurIPS. [arXiv:2311.17238]
- K. Napier, ..., **M. Zhao** et al. "COOL-LAMPS. Discovery of COOL J0335-1927, a Gravitationally Lensed Quasar at $z = 3.27$ with an Image Separation of 23.3." *The Astrophysical Journal Letters*, vol. 954, no. 2, 1 Sept. 2023, pp. L38–L38, [arXiv:2305.14317].

RESEARCH EXPERIENCE

Ph.D. thesis research Aug, 2024 - Present
Statistical & Computational Cosmology (advised by Bhuvnesh Jain & Gary Bernstein) Philadelphia, PA

- Develop higher-order statistics pipeline for Dark Energy Survey Year 6 data
- Develop field-level inference pipeline, focusing on extracting the relation between galaxies and matter density

B.S. Honors thesis research June, 2023 - June, 2024
Machine learning in Supernova Cosmology (advised by Joshua Frieman) Chicago, IL

- Developed machine learning algorithm to predict distance modulus from simulated SNIa dataset using Convolutional Neural Network (CNN)
- Implemented novel data processing method to turn light curve data into image data suitable for ML training

COOL-LAMPS collaboration, UChicago Department of Astronomy & Astrophysics Jan, 2023 - June, 2023
Observation and Analysis of Lensed Quasars (advised by Michael Gladders) Chicago, IL

- Observed potential wide-separation lensed quasars using the Magellan Telescopes for 9 nights
- Performed SED analysis and confirmed the discovery of the newest wide-separation lensed quasar
- Performed lens and source modeling on both Magellan and Herschel data using Galfit

Deep Skies Lab, Fermi National Accelerator Laboratory June, 2022 - June, 2023
Domain Adaptation for Strong Lensing (advised by Brian Nord & Aleksandra Ciprijanovic) Batavia, IL

- Led and developed a machine learning algorithm using a domain adaptation technique called Maximum Mean Discrepancy to estimate the Einstein radius on simulated strong lensing data
- Designed and simulated strong lensing data imitating the Dark Energy Survey

AWARDS

- Dean's Fund for Undergraduate Research Conference, \$900
- The Quad Undergraduate Summer Research Grant, \$5500

- Jeff Metcalf Fellowship $\times 2$, \$950 $\times 2$

PRESENTATIONS

- Poster: 243rd Meeting of the American Astronomical Society Jan, 2024
- Poster: NeurIPS, The Machine Learning and the Physical Sciences workshop Dec, 2023
- Poster: Debating the potential of machine learning in astronomical surveys (II) Nov, 2023
- Poster: Midstates Undergraduate Research Symposium in Physical Sciences, Math and Computer Sciences Nov, 2023
- Talk: 2023 Fermilab New Perspectives June, 2023
- Poster: 2023 UChicago Undergraduate Research Symposium Apr, 2023
- Poster: 2023 Conference for Undergraduate Women in Physics Jan, 2023

TEACHING & MENTORING

Mentor for First-year Graduate Student

Mentee: Benjamin Du Pont

Aug, 2025 - Present

Philadelphia, PA

- help first-year graduate student navigate coursework and research

Teaching Assistant

Instructor: Robert R Johnson

Aug, 2024 - June, 2025

Philadelphia, PA

- PHYS 0150 Principles of Physics I: Mechanics and Wave Motion: lead laboratory exercises
- PHYS 0151 Principles of Physics II: Electromagnetism and Radiation: lead in-class exercises and hold office hours

Undergraduate Teaching Assistant

Instructor: Andrey Kravtsov

Jan, 2024 - Mar, 2024

Chicago, IL

- ASTR 21100 Computational Techniques in Astrophysics: hold office hours, grade homework, and lead final projects

LEADERSHIP & OUTREACH

Diversity & Inclusion in Physics (DIP) at Penn

Co-president

Aug, 2025 - present

- build safe spaces for discussions on department climate and communicate with department leadership
- hold reading groups and speaker events on experiences of marginalized groups in physics

Society of Asian Scientists and Engineers (SASE) at UChicago

Social Chair

Jan, 2023 - June, 2024

- Organized campus wide social events that facilitate connections among the Asian science community
- Founding member and assisted in setting up bylaws and logistics

Society of Women in Physics (SWiP) at UChicago

Co-outreach Chair & Webmaster

Sep, 2022 - June, 2024

- Co-organized speaker events, mentorship program, and meeting with department officials
- Maintained the official SWiP website and Instagram page; compiled research, funding, and internship resources

REFERENCES

- **Joshua Frieman** Menlo Park, CA
SLAC National Accelerator Laboratory jfrieman@slac.stanford.edu

- **Michael Gladders**
University of Chicago

Chicago, IL
gladders@astro.uchicago.edu

- **Aleksandra Ciprijanovic**
Fermi National Accelerator Laboratory

Batavia, IL
aleksand@fnal.gov